

A Novel Image-Graph Heterogeneous Fusion Framework for Static IR Drop Prediction

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Abstract—IR drop analysis is crucial for ensuring the reliability and performance of integrated circuits (ICs) but poses computational challenges as the IC designs grow larger, especially for ultra deep-submicron VLSI designs. Deep learnings (DL) as the efficiency-promising solutions, mainly employ various CNN-based networks to achieve image-to-image IR drop predictions. However, they neglect and lose the power delivery network (PDN) global spatial features and cell instance topological information. This paper proposes a novel image-graph heterogeneous fusion framework (IGHF), which integrates the effectiveness and complementarity of dual branches (CNN and GNN) for higher prediction performance. In the CNN-based Power ScaleFusion Unet branch, the proposed long-range and local-detail encoder (LLE) integrates seamlessly with the hierarchical and adjacent compensation group (HACG) module. This design facilitates effective multi-scale global-to-local spatial power feature extraction within the PDN and enables adaptive high-to-low-level feature fusion and compensation in the decoder. Moreover, a cell voltage aware (CVA) module in the GNN branch is designed to adaptively aggregate PDN topological features of heterogeneous neighbors of different orders. Comparative experiments demonstrate that the proposed IGHF achieves significant accuracy improvements, outperforming the state-of-the-art MAUNet and widely-used IREDGe methods by considerable margins of 24.6% and 55.0% reduction in prediction error, while the prediction maps possess higher structural fidelity. Transfer experiments indicate that IGHF with transfer learning can improve the accuracy in real circuits with the few-shot real circuit test cases.

Index Terms—IR drop prediction, Convolutional neural network, Graph neural network, Heterogeneous fusion.

I. INTRODUCTION

IR drop analysis refers to obtaining the voltage drops caused by the supply current flowing through the power delivery network (PDN) [1]. The demand for IR drop analysis is gradually increasing due to the lower noise margin required for power reliability in low-power designs. It is crucial due to its close relation to chip performance, reliability, and functionality [2]. As the chip integration density escalates, traditional simulation-based IR drop analysis requires formidable computational and memory costs owing to high-dimensional circuit equations [3], [4]. Moreover, various methods including

the multi-grid method [5], the boundary element method [6], and the random walk algorithm [7] are designed to mitigate the computational burden. However, the computational overhead of matrix solution operations remains substantial, especially for irregular and non-uniform power grids [8].

Currently, convolutional neural network (CNN)-based deep learning methods provide efficient and promising solutions for full-chip IR drop prediction [9]–[13]. IREDGe [12] introduces the pad distribution feature and employs encoder-decoder architecture for pixel-level prediction. In [13], MAUNet integrates multi-scale convolutional blocks, attention mechanisms, and U-Net to optimize IR drop prediction accuracy. However, due to the constraints imposed by the limited size of convolution kernels, these approaches primarily emphasize the IR drop effects induced by local-area cell instances, making it challenging to capture the global IR drop patterns, particularly the distribution of cell current and resistance [14], [15]. The sizes of the kernels for spatial feature extraction in CNN pose challenges. Small windows risk violating the locality principle of IR drops, whereas larger windows for the global receptive field are information-leaking, memory-intensive, and time-consuming [16]. In this case, *how to improve CNNs to capture both the global and local PDN spatial features for higher IR drop prediction accuracy is vital and under exploration.*

In addition, not only do the current/instance load patterns impact the IR drop distribution, but the PDN topologies also contain the actual cell current flow and interference information, which are indispensable for IR drop [9], [17]. However, CNNs primarily capture image-like representations from current load patterns. While they lack topology perception mechanisms [18], and a single image pixel may contain multiple cell instances, which loses instance information that affects the hotspot area [19]. Therefore, it's a challenge to represent the practical non-Euclidean PDN connectivities effectively [20]. It's essential to introduce rich and supplementary PDN topological features to address the problem of inconsistent correspondence between images and cell instances. Currently, graph neural networks (GNNs) have been widely applied to

capture the circuit topology, addressing issues such as routing congestion and power performance area (PPA) optimization [21], [22]. However, *how to introduce GNN to effectively incorporate the PDN topological features for higher IR drop prediction performance remains a novel challenge.*

This work proposes an image-graph heterogeneous fusion framework (IGHF) that integrates a CNN-based branch with a GNN-based branch. The CNN-based branch employs a Power ScaleFusion Unet, while the GNN-based branch constructs a cell voltage aware module. The main contributions include:

- A novel image-graph heterogeneous fusion framework (IGHF) is proposed, which integrates a CNN-based branch to capture multi-scale global-to-local PDN spatial features and a GNN-based branch to perceive the non-Euclidean connection relation in the PDN topology.
- In the CNN branch, a long-range and local-detail encoder (LLE) is proposed to integrate the long-distance regional superposition effects and local instance-specific current characteristics. Moreover, a hierarchical and adjacent compensation group (HACG) module is designed to adaptively rescale the multi-scale PDN spatial features for guiding a high-to-low-level feature fusion.
- In the GNN branch, the cell heterogeneous edge construction is designed to ease information propagation. Moreover, a cell voltage aware (CVA) module is proposed to adaptively aggregate PDN topological features from heterogeneous neighbor cells of different orders.

II. PRELIMINARIES

IR drop analysis aims to compute the voltage drops induced by the supply current flowing through the PDN. It is obvious that both the current load patterns and the PDN architectures affect analysis accuracy [23]. The precise IR drop analysis is crucial since voltage overshoot can lead to irreversible device damage and chip malfunction.

For the full-chip IR drop analysis, a prevailing method in commercial software is Modified Nodal Analysis (MNA), where the Kirchhoff Current Law is employed to formulate a system of linear equations. The system is characterized by a conductance matrix G for the PDN, a node voltage vector V , and a current source vector J , represented by $GV = J$ [24].

However, the system can rapidly balloon to millions of voltage nodes and interconnect segments in modern chip designs as the integration density escalates, which makes the analysis both computationally intensive and memory-demanding [25]. Therefore, machine learning-based IR drop predictions are introduced to accelerate the process.

III. THE PROPOSED METHOD

In this work, our DL-driven IR drop prediction method employs a heterogeneous dual-branch network, as shown in Fig. 1. It integrates a CNN-based branch for multi-scale global-to-local spatial power feature extraction and a GNN-based branch to complement PDN topological information, which significantly boosts the prediction performance.

A. Input Image and Graph Construction

Image Construction: The inputs of the CNN branch are the feature images I according to the mapping of spatial cell coordinates, which are extracted from raw layout and netlist files. To enhance analysis convergence and reduce redundant features, the Permutation Importance (PI) filter proposed in [26] is employed. This method randomly permutes the order of the original input images I and identifies features as critical for the analysis if they result in significant changes in model accuracy upon permutation, with a correlation coefficient importance $\geq \alpha$. In this work, the power features (total power, leakage power), representing current spatial distribution, and the resistance feature (min. path resistance), representing the worst-case PDN current condition, are employed. In contrast, we exclude toggle power, internal power, switching power, effective path resistance, and cell frequency.

Graph Construction: An unweighted bidirectional heterogeneous graph $G = (V, E, R)$ is constructed based on the .dspf files to represent the practical non-Euclidean PDN topology, as illustrated in Fig. 1. In this graph, the set V comprises all cell instances v , while the set E represents the connections e between them. The set R encompasses all types of relationships between v (the following R_I and R_d). The above-selected power and resistance features are assigned to V 's attributes. Next, heterogeneous edge relations are constructed according to two criteria. The first is the original PDN instance connection relation R_I , representing the voltage propagation by the practical current flowing. The second is the designed cell distance-aware edge relation R_d , which adds the physical-inexistent edges considering the spatial adjacencies (the Euclidean distance $\|v_i, v_j\|_2 \leq \delta$ for the instances v_i, v_j). The virtual edges ease the information propagation in GNN. Thus, R consists of relations R_I and R_d . The heterogeneous graph G and outputs of CNN are used as the input to GNN.

B. CNN Branch: Power ScaleFusion Unet

The CNN branch adopts the Unet architecture. First, the long-range and local-detail encoder is proposed to achieve multi-scale global-to-local spatial power feature extraction. In decoder, the hierarchical interaction compensation group is designed to adaptively rescale and guide hierarchical feature fusion and compensation.

1) Long-Range and Local-Detail Encoder (LLE): Due to the localized IR drop phenomena caused by nearby cells and the global IR drop phenomena resulting from regional activity, IR drop at each location relies on both local and global power information. Therefore, the LLE, combining long-range modules and local-detail modules, is proposed to capture the multi-scale long-distance and local-detail PDN power and current load characteristics, as shown in Fig. 1.

Long-Range Module (LRM): The LRM in LLE serves as a global PDN extractor, which achieves the long-distance spatial power characteristic awareness through the long-range block (LRB), as shown in Fig. 2 (a). For LRB, inspired by the Swin Transformer [27], it is divided into a perception network and a convolutional feed-forward network (FFN). In the perception

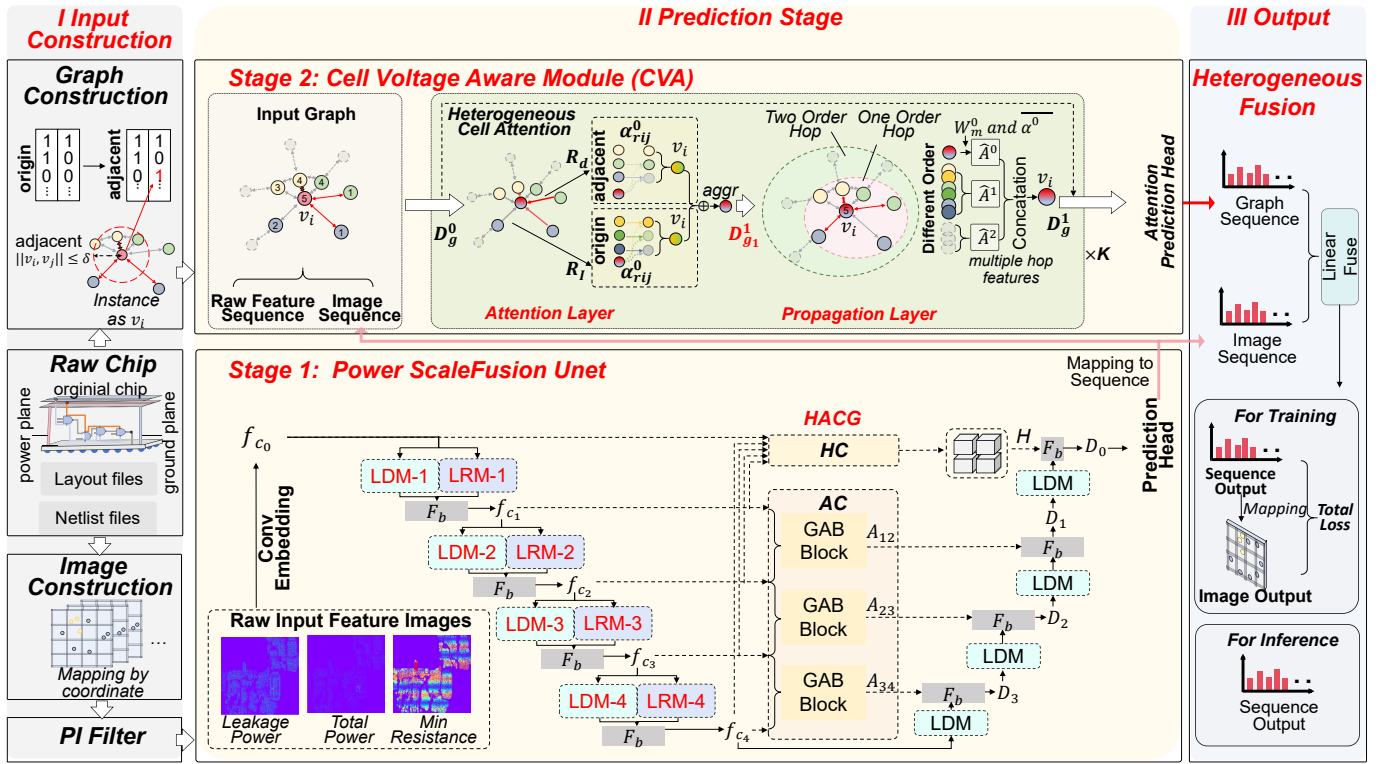


Fig. 1: Proposed IGHF framework for IR drop prediction. First, raw SoCs are transformed into images and graphs in the left part. Then the GNN-based branch (the upper in middle part) and CNN-based branch are proposed to extract the PDN topological features and multi-scale global-to-local spatial features from the graphs and images, respectively. Finally, the aligned GNN features and the CNN features are fused to generate the final prediction in the right part.

network, the large kernel depthwise (DW) convolution [28] is introduced to significantly improve the effective receptive fields to contain broader power and current distribution information. The design of large convolution kernels, as opposed to attention mechanisms in the Swin Transformer, is better suited for addressing problems with limited circuit data due to their inherent inductive bias. Additionally, large kernels help mitigate optimization challenges associated with increasing model depth when relying on stacked small kernels.

Furthermore, to mitigate the memory-intensive costs of LRB due to the large kernel design, LRB is designed as an inverted bottleneck residual structure as a whole [29]. The inefficient large-kernel convolutions are designed with fewer channels, whereas the more efficient and dense 1×1 layers within the FFN are responsible for managing the heavy computational workload, including feature dimensionality and the model's non-linearity. Moreover, multi-level LRMs allow model to perceive various-scale spatial power characteristics.

Local-Detail Module (LDM): Different from the LRM for long-distance power features, the LDM with small kernel convolutions focuses on encoding fine-grained PDN spatial features and obtaining distinct small-scale current load characteristics of standard cells. As shown in Fig. 2 (b), it employs more non-linear Leaky ReLU (LReLU) to enhance encoding diversity. Moreover, the stacking LDMs further extract multi-scale local-detail power features.

Branch Fusion F_b : The long-range dependencies and fine-grained information from each LRM and LDM are superposed by the branch fusion layer. In this work, the small kernel convolution operation is chosen as the fusion operation F_b . The fusion layer is updated by Eq. (1), where $f_{c_i} (i = 1, \dots, 4)$ denotes the encoder feature map.

$$f_{c_i} = F_b\{LDM-i(f_{c_{i-1}}), LRM-i(f_{i-1})\} \quad (1)$$

2) **Hierarchical and Adjacent Compensation Group (HACG):** The multi-scale power encoder features contribute differently to IR drop due to significant variations in PDN size and cell power density [30], [31]. Different from the simple skip connections in vanilla Unets, the HACG in the decoder is proposed. It combines a hierarchical compensation and adjacent compensation module to achieve the different-scale encoder feature map fusion interactions and compensate the decoder information degradation through the lightweight group aggregation bridge (GAB), as shown in Fig. 3. It enables the fusion of multi-scale information at the final decoding stage and reduces sampling losses at different scales during the upsampling process.

Hierarchical Compensation: First, the hierarchical compensation (HC) is designed for the top decoder layer to integrate feature maps with different semantics at each scale. Owing to the varying receptive fields, it is necessary to superimpose hierarchical features to model the trend of voltage

fluctuations more comprehensively. The top decoder combines the output of the top encoder with those of the other encoders through multiple dilated convolution groups within the HC module. This design effectively leverages the contributions of different-scale encoder semantics to enhance the representation of node voltages. The HC output H is computed as Eq. (2).

$$H = HC(f_{c_0}, \dots, f_{c_N}) = \sum_{i=1:N} GAB(f_{c_0}, f_{c_i}) \quad (2)$$

where N denotes the total number of high-level encoded features to integrate. $GAB(f_{c_0}, f_{c_i})$ utilizes depthwise separable convolution and bilinear interpolation to align low-level and high-level feature sizes f_{c_0} and f_{c_i} and utilizes four different dilation convolution groups to extract information adaptively.

Adjacent Compensation: The adjacent compensation (AC) module primarily aims to compensate for the inevitable local information degradation of the downsampling layers. $A_{i,i+1} = GAB(f_{c_i}, f_{c_{i+1}})$ performs GAB interaction on adjacent encoder maps $f_{c_i}, f_{c_{i+1}}$ to adaptively rescale and compensate the corresponding decoder layer D_{c_i} , which guides a high-to-low level IR drop prediction. Different from the standard skipping connections between the corresponding encoder and decoder layers, AC reduces the upsampling loss of features between adjacent layers.

C. GNN Branch: Cell Voltage Aware Module (CVA)

The GNN-based branch utilizes the CVA for PDN topology feature extraction to consider the intricate topological connections and adjacent effects. CVA consists of cascade K residual blocks, the block is composed of an attention layer and a propagation layer. The CVA adaptively learns the influence weights of heterogeneous connections from neighboring nodes to the central node in the attention layer. Moreover, it combines voltage-related features from neighbors at varying distances in the propagation layer, as shown in Fig. 1.

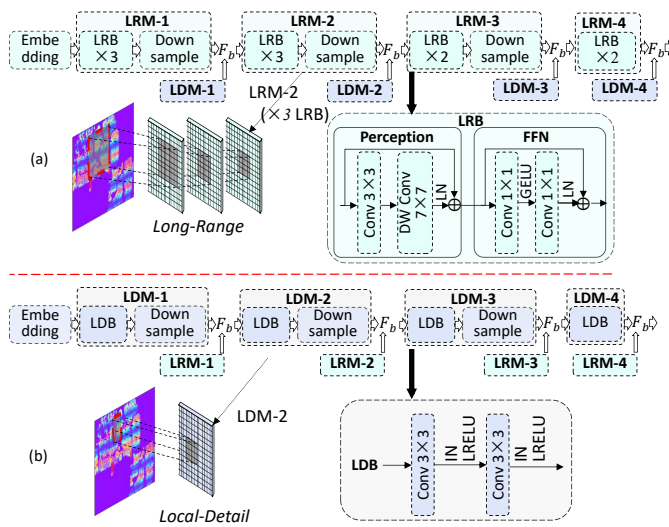


Fig. 2: The architecture of LLE. The bottom left corner contrasts the receptive field sizes for LRM (a) and LDM (b). The rest is the detailed LRM and LDM. Here L is 1.

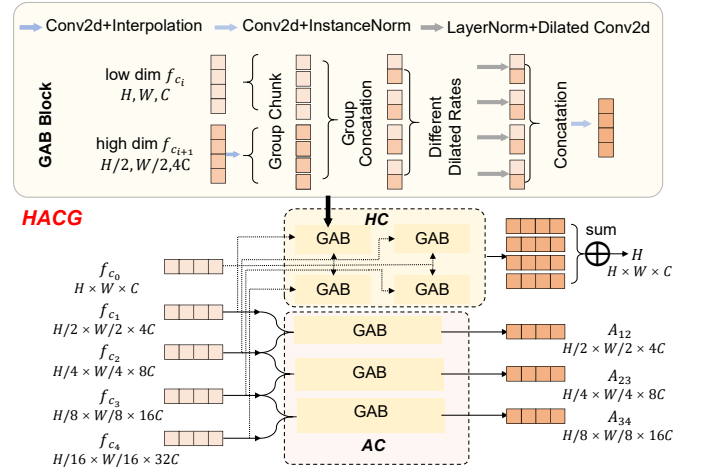


Fig. 3: The architecture of the HACG. The top yellow part denotes the GAB block. And the rest is the HACG consisting of HC and AC submodules. H, W, C represent the height, width, and number of channels of the features, respectively. Here dilated rates of convolution groups are 1, 2, 5, and 7.

Attention Layer: The virtual edge and the actual current transmission edge exhibit heterogeneous characteristics in their impact on the voltage of the central cell node. And the cell instance connection edges E cause complex interactive relationships due to discriminative distances or resistances between nodes. To adaptively rescale and aggregate significant information from heterogeneous edge connections, the heterogeneous cell attention mechanism is designed as Eqs. (3) and (4). It learns the critical interactions between nodes in heterogeneous edge type r , using the attention factor α_{rij}^l . Then all outputs are aggregated into $D_{g_1}^l$. Moreover, the average of α_{rij}^l is also used as the attention factor of propagation layer for two heterogeneous edge connection weights.

$$D_{g_1}^l = \sum_r LReLU(\alpha_r^l \circ (\hat{W}_r^l A_r D_g^l)) \quad (3)$$

$$\alpha_{rij}^l = \text{softmax}\left(\frac{\exp(e_{rij}^l)}{\sum_{j \in N_i} \exp(e_{rij}^l)}\right) \quad (4)$$

where D_g^l and $D_{g_1}^l$ is the input and output of the l_{th} -layer attention layer, r denotes the edge type in the relation set R , $\hat{W}_r^l$ is the linear parameter in type r , N_i represents neighbor set of v_i . α_{rij}^l as an element of matrix α_r^l , is the attention weight of v_i to v_j in type r . e_{rij}^l is the attention score between v_i and v_j in type r , using Eq. (2) of [32].

Propagation Layer: The voltage fluctuation of a node v_i is considered as the superposition of multiple surrounding cell nodes. This layer learns voltage representation through adjacent virtual and practical physical connections from neighbors. Different-order neighbors influence the central node to varying degrees. Vanilla graph convolutional networks (GCNs) only aggregate first-order neighborhood information, neglecting higher-order neighborhood data [33]. Consequently, these methods are suitable only for graphs where the first-order

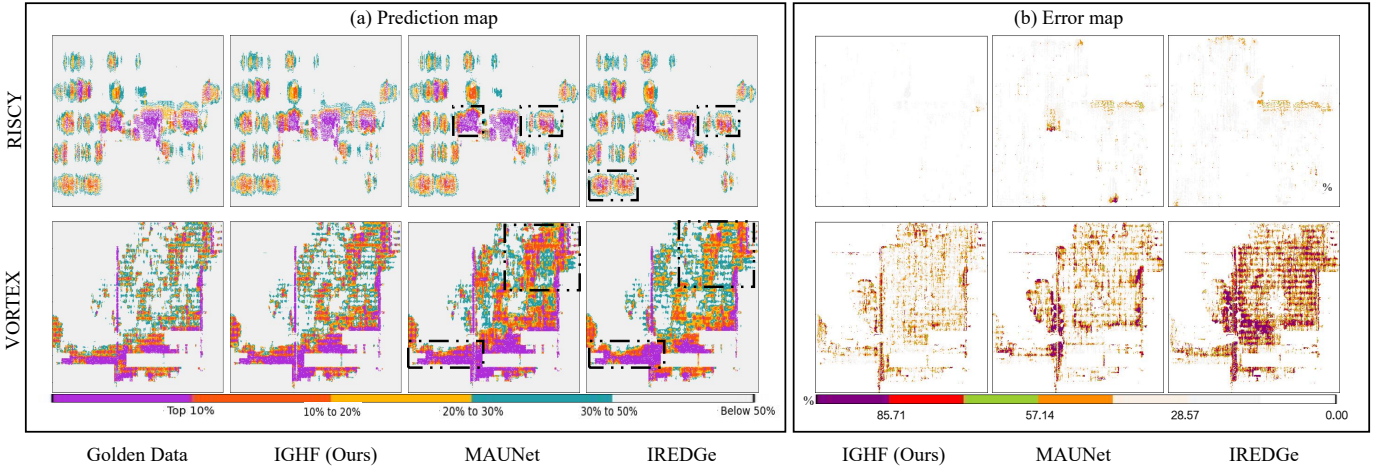


Fig. 4: two visual cases for RISCY (44,855 cell instances for CPU chip) and VORTEX (108,028 cell instances for GPU chip). (a) Prediction maps: The top 10% and 20% of the IR drops are highlighted as purple and red hotspots. The top 30% and 50% of the IR drops are indicated as yellow and green regions. (b) Relative error maps between golden data and two predictions.

neighbors exhibit high compatibility with the central node v_i . This limitation renders them ineffective for graphs where the first-order neighbors are incompatible with the central node v_i , but higher-order neighborhoods demonstrate compatibility.

To address this, MixHop is employed [33]. It learns the relationships among different cells by repeatedly mixing the voltage-related features of neighbors at varying distances and incorporating different operators. In this approach, graph convolution as described in Eq. (5) can represent the information aggregation of neighbors spanning two order hops. MixHop's node update method can extract linear combinations of neighbor features from a farther distance, allowing it to capture meaningful signals even in extremely sparse connections. Eq. (5) aggregates the information of self-loops, first-order and second-order neighbors through the average attention factor α^l in attention layer and its own learnable weight W_m^l to form the output of the l -th layer, where $m = 0, 1, 2$.

$$D_g^{l+1} = LReLU \left(\left[\bar{\alpha}^l I D_{g_1}^l W_0^l \mid \bar{\alpha}^l \hat{A} D_{g_1}^l W_1^l \mid \bar{\alpha}^l \hat{A}^2 D_{g_1}^l W_2^l \right] \right) \quad (5)$$

where $\hat{A}$ is a adjacency matrix with self-connections, $D_{g_1}^l$ and D_g^l is the input and output of the l -th layer, I is the diagonal degree matrix, W_m^l is the learnable weight. $\mid$ denotes concatenation. And after passing through K residual blocks, the graph instances are processed by the GAT prediction head to produce the final graph sequence.

Fusion: To generate the final IR drop prediction sequence, the final graph sequence is linearly fused with the CNN image output, leveraging the node image sequence aligned with the cell instance coordinates.

IV. EXPERIMENTS AND RESULTS

A. Experiments Setting and Results

The experiment was conducted on an Ubuntu computer with a GeForce RTX 3090 GPU, using the public CircuitNet-ICISC benchmark dataset. This dataset includes over 7,000 samples

from five 28-nm open designs, each with varying layouts, 54 synthesized circuit netlists (with different timing constraints and macros), and cell utilizations [34]. The training to test set ratio was 8:2.

We use an L1/L2 loss with Adam optimizer, including a 5-epoch linear warm-up and cosine annealing (1.5×10^{-3} to 1×10^{-5}). The CNN is pre-trained on 320×320 images for 100 epochs, then frozen and combined with the GNN and fusion block. During training, the model outputs instance sequences and images, with the total loss calculated as the sum of outputs for supervised learning. For IGHF's filter settings: stage 1 (CNN) has embedding [3,32], LLE [32,64,128,256,512], and decoder [512,256,128,64,32,1]; stage 2 (GNN) uses CVA [4,16,32,64,128,64,32,16,1]; the heterogeneous fusion stage has linear fuse [2,1].

To evaluate the performance, the proposed IGHF is compared with two excellent CNN-based methods including newly-proposed SOTA MAUNet [13], and widely-used IREDGe [12]. IGHF, MAUNet, and IREDGe have training times of 12, 8, and 5.5 hours, respectively. Note that, the famous method PowerNet [11] is excluded due to its unaffordable forward time. Moreover, accuracy and graph metrics (Pearson correlation coefficient (PCC), coefficient of determination R^2 , mean absolute error (MAE) (mV), and average max MAE (AMMAE) (mV)) are used for assessment [34]. MAE evaluates prediction accuracy, AMMAE assesses errors in hotspot regions, and PCC and R^2 measure precision.

From Tab. I, IGHF shows significant performance gains in all four metrics. Compared to MAUNet, our method still outperforms with a notable margin in metrics (MAE: **0.378 mV** vs. 0.501 mV (over **24.6%** ↓) PCC: **0.973** vs. 0.961). It also reduces MAE by 55.0% compared to IREDGe. Moreover, lower AMMAE indicates the better robustness of IGHF in worst-case situations. In addition, Fig. 4 shows prediction and error maps for two chips. Compared with the SOTA MAUNet and IREDGe methods, our prediction images have

TABLE I
PREDICTION PERFORMANCE COMPARISONS.

RTL	IGHF (Ours)				IREdGe [12]				MAUNet [13]			
	PCC $\uparrow$	R^2 $\uparrow$	MAE $\downarrow$	AMMAE $\downarrow$	PCC $\uparrow$	R^2 $\uparrow$	MAE $\downarrow$	AMMAE $\downarrow$	PCC $\uparrow$	R^2 $\uparrow$	MAE $\downarrow$	AMMAE $\downarrow$
ZERO-RISCY	0.975	0.934	0.360	17.818	0.932	0.808	0.793	32.250	0.960	0.897	0.489	24.944
RISCY	0.974	0.940	0.357	22.123	0.930	0.790	0.790	41.513	0.961	0.899	0.473	28.012
RISCY-FPU	0.977	0.941	0.358	21.839	0.943	0.750	0.814	38.061	0.966	0.897	0.486	27.386
VORTEX	0.908	0.791	1.310	267.136	0.882	0.771	2.054	392.454	0.899	0.780	1.430	269.440
NVDLA	0.863	0.742	1.311	196.230	0.704	0.567	5.750	192.005	0.849	0.731	1.374	199.104
Total	0.973	0.910	0.378	24.790	0.933	0.795	0.840	41.296	0.961	0.874	0.501	30.988

TABLE II
TRANSFER PERFORMANCE COMPARISONS.

RTL	Model	PCC	R^2	MAE	AMMAE
VORTEX Avg. Cells: 109944	IGHF	0.891	0.783	2.223	251.098
	IGHF w/ TL	0.938	0.859	1.053	183.207
	MAUNet	0.888	0.721	1.340	288.057
	MAUNet w/ TL	0.919	0.756	1.113	254.213
NVDLA Avg. Cells: 279228	IGHF	0.803	0.718	2.223	225.207
	IGHF w/ TL	0.908	0.811	0.919	182.632
	MAUNet	0.737	0.663	3.837	244.402
	MAUNet w/ TL	0.880	0.753	1.261	182.791

higher structural fidelity. There are significant error reductions in multi-level (high and low IR drop regions, especially in the top 10% and 20% drop hotspot areas) and multi-scale (global and local) IR drop regions.

B. Transfer Experiments

To adapt to different IR drop characteristics of circuits, transfer learning (TL) is used to enhance the IGHF model's performance and generalization for limited unseen real-circuit test cases. To adapt to different PDN designs, we applied the same fine-tuning scheme, scale and shift feature adapter (SSF-ADA)[35], as MAUNet. The pre-trained set includes ZERO-RISCY, FPU, and RISCY-FPU RTL (avg. cell count: 46,290). The fine-tuning set has some NVDLA and VORTEX design cases (avg. cell counts: 109,944 and 279,228), while the remainings are kept hidden for evaluation. During training, the encoder was frozen, while the SSF-ADA module and decoder were activated. The IGHF model's performance with MAUNet was compared before and after TL in Tab. II.

The IGHF model consistently outperforms the MAUNet model in Tab. II, both before and after TL. Specifically for the NVDLA RTL, the MAE of IGHF with TL is reduced by 58.70% compared to IGHF without TL (from 2.223 mV to 0.919 mV). Furthermore, the MAE of IGHF with TL decreases by 27.11% compared to that of MAUNet with TL.

C. Ablation Experiments

This section performs two ablation experiments. First, we investigate the effectiveness of heterogeneous feature fusion by comparing line 1 (Proposed IGHF), line 2 (IGHF without GNN), and line 3 (IGHF without CNN) in Tab. III. The results show that combining the improved CNN branch with the GNN branch significantly enhances IR drop prediction performance,

TABLE III
ABLATION EXPERIMENTS FOR IGHF.

GNN		CNN		Metrics			
CVA	LLE	HACG		PCC $\uparrow$	MAE $\downarrow$	R^2 $\uparrow$	AMMAE $\downarrow$
$\checkmark$	$\checkmark$	$\checkmark$		0.973	0.378	0.910	24.790
	$\checkmark$	$\checkmark$		0.969	0.389	0.905	27.717
$\checkmark$				0.832	1.171	0.781	41.790
	$\checkmark$			0.964	0.456	0.897	29.070
		$\checkmark$		0.958	0.458	0.895	29.037

as the GNN branch supplements the PDN topological awareness that the CNN branch lacks.

For the CNN branch, the effectiveness of the proposed LLE and HACG modules can be assessed by comparing line 2 with lines 4 and 5, respectively. It is clear that both the LLE, which extracts multi-scale global-to-local spatial power features, and the HACG module, which facilitates different-scale feature adaptive interaction and compensation, are beneficial for significantly enhancing prediction accuracy.

V. CONCLUSIONS

In this paper, a novel image-graph heterogeneous fusion framework is proposed, which combines a CNN branch for the multi-scale global-to-local spatial feature extraction, and a GNN branch for aggregating discriminative cell instance topological features. In the CNN branch, LLE and HACG modules complete the global and local PDN feature perception and the fusion of features at each scale, respectively. In the GNN branch, the attention and propagation layers in CVA adaptively aggregate PDN topological features of heterogeneous neighbors of different orders. Extensive experiments on a large public dataset demonstrate that our IGHF outperforms the newly-proposed SOTA MAUNet and popular IREDGe methods for IR drop prediction by a large margin (over 24.6% and 55.0%). Transfer experiments denote that the error metrics of IGHF with TL show a decrease of 27.1% compared to those of MAUNet with TL. We hope that the proposed method can advance the development of CNN-GNN heterogeneous architecture design in the IR drop analysis.

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